

intro

APPRENTICESHIP PORTAL

The PWA Portal is a dedicated digital platform built by Even Cargo to address the chronic underrepresentation of women in India's apprenticeship ecosystem. With women constituting only 21.9% of total apprentices nationally and female enrolment declining 41% between FY 2021-22 and FY 2023-24, a technology-driven intervention is not optional - it is urgent.

The portal will serve as the single operating system connecting 3 key stakeholders: women apprentice candidates, employer partners, and the Even Cargo admin team.

It will manage the full apprenticeship lifecycle - from awareness and sourcing, through onboarding, NAPS compliance, stipend processing, and post-placement tracking.

candidate use case

ID	Use Case	Actor	Description
UC-C01	Self-Registration	Candidate	Candidate registers via mobile using phone number + OTP. Completes profile in her preferred language with guided prompts and voice input support.
UC-C02	Profile & Document Upload	Candidate	Uploads Aadhaar, education certificates. OCR auto-fills form fields. Documents stored securely.
UC-C03	Apprenticeship Discovery	Candidate	Browses available apprenticeships filtered by location, sector, stipend range, duration, and qualification requirement.
UC-C04	Application Submission	Candidate	Applies to one or more apprenticeship opportunities with a single tap. Tracks application status in real time.
UC-C05	Interview Scheduling	Candidate	Confirms, reschedules, or requests virtual interview option.
UC-C07	Onboarding Checklist	Candidate	Completes onboarding tasks: NAPS registration.
UC-C08	Training Tracker	Candidate	Tracks her training progress, attendance log, skill milestones, and upcoming assessment dates.
UC-C09	Stipend Tracking	Candidate	Views stipend payment history, expected payment date, and bank transfer status for each pay cycle.
UC-C10	Grievance Submission	Candidate	Raises a workplace grievance (safety, non-payment, harassment) through a confidential, single-click flow. Status tracked in app.
UC-C11	Chatbot Support	Candidate	Accesses a multilingual chatbot for FAQs, document guidance, and escalation to a human agent.

Candidate Portal Pages

Candidate Dashboard (Home)

- Personalised greeting with name and profile completion meter
- Application status cards (Applied / Interview Scheduled / Offer Received / Active)
- Upcoming events: interview dates, assessment dates, stipend due dates
- Quick actions: Browse Jobs, Track Application, Raise Issue, SOS
- Financial literacy module progress tracker
- Community feed snippet (3 stories)

Profile & Documents

- Personal information: name, DOB, address, contact, emergency contact
- Education details: qualification, institution, year of passing

- Document vault: Aadhaar, education certificate, bank passbook, photo — upload, preview, replace
- NAPS profile card (auto-generated once NAPS registration complete): Apprentice ID, trade, employer
- Privacy controls: what data is visible to employers vs. facilitators

Job Discovery & Search

- Search bar + advanced filters (same as public explorer, but personalised by profile)
- 'Recommended for you' section based on qualification and location
- 'New in your area' section based on GPS / registered address
- Map view of nearby opportunities
- Each listing: full details including exact workplace address, working hours, safety rating, employer gender policy, number of women already placed there

Application Tracker

- Timeline view of each application: Applied → Shortlisted → Interview → Offer → Active / Rejected
- For each step: date, action required, contact person, document needed
- Offer letter preview and digital acceptance flow
- Interview prep tips (text + video, in language)

My Apprenticeship (Active Programme Tracker)

- Current placement details: employer, role, location, supervisor name and contact
- Attendance calendar: marked by employer supervisor, viewable by candidate
- Training progress: skills/modules completed, pending assessments
- Assessment schedule and results
- Duration tracker: days completed vs. total contract duration

Stipend & Payments

- Payment history table: month, amount, status (Paid / Pending / Processing)
- Next expected payment date
- Bank account details (view-only, edit via document re-upload)
- Flag a payment issue button (goes to grievance flow)

Grievance & Support

- Grievance categories: Workplace Safety, Non-Payment, Harassment, Discrimination, Poor Training Quality, Other
- Confidential mode (employer is not informed of identity during investigation)
- Grievance status tracker: Submitted → Under Review → Resolved / Escalated

employer use case

ID	Use Case	Actor	Description
UC-E01	Employer Registration	Employer	Company registers with CIN, GST, and authorized signatory details. Verified by Even Cargo admin before activation.
UC-E02	Job Posting Creation	Employer	Creates apprenticeship listings specifying trade, qualification, stipend, location, duration, number of seats, and working hours. Even Cargo reviews for NAPS alignment before publishing.
UC-E03	Candidate Pool Browsing	Employer	Browses a pre-screened pool of women candidates filtered by qualification, location, trade readiness, and availability.
UC-E04	Interview Management	Employer	Schedules interviews, sends automated reminders, marks attendance, and records outcomes through the platform.
UC-E05	NAPS Registration Management	Employer	All NAPS portal filings handled by Even Cargo on employer's behalf. Employer views filing status and gets confirmation receipts.
UC-E06	Attendance & Training Logs	Employer	Supervisor marks attendance and training milestones. System flags anomalies (e.g., sustained absences) for follow-up.
UC-E07	Stipend Processing	Employer	Uploads monthly stipend payment confirmation or initiates payment via integrated payment gateway. System validates NAPS-mandated stipend amounts.
UC-E08	Compliance Dashboard	Employer	Single-view dashboard showing all active apprentices, their NAPS status, documentation completion, and upcoming deadlines.
UC-E09	Government Subsidy Tracking	Employer	Tracks NAPS reimbursement claims (up to Rs. 1500/month per apprentice) filed by Even Cargo. Views disbursement status.
UC-E10	ESG Reporting Export	Employer	Downloads apprenticeship data formatted for ESG reporting — number of women hired, sectors, stipends paid, certificates issued.
UC-E12	Grievance Response	Employer	Receives and responds to candidate grievances through the platform. Resolution tracked and time-bound.
UC-E12	Multi-Location Management	Employer	Manages apprenticeship listings and apprentice cohorts across multiple work locations from a single account.

Employer Portal Pages

Employer Dashboard

- KPI cards: Active Apprentices, Open Positions, NAPS Compliance Score, Pending Actions, Stipends Due
- Alert centre: document expiry, stipend due, apprentice milestone, government subsidy status
- Quick actions: Post New Listing, View Applications, Download ESG Report
- Upcoming interview schedule

Company Profile

- Company details: name, CIN, registered address, GST, industry sector
- Gender policy declaration: maternity policy, POSH compliance, safety certification
- Employer safety score (computed by Even Cargo, visible to candidates)
- Upload: POSH policy, safety audit certificates, ESG commitment letter
- Authorised signatory management

Job Postings Management

- Create new listing: trade, qualification required, stipend offered, seats, location, working hours, start date
- NAPS compliance checker (real-time): flags any field that doesn't meet NAPS norms
- Listing status: Draft / Under Review / Live / Paused / Closed
- Performance metrics per listing: views, applications received, shortlisted, offered, filled

Candidate Management

- Application inbox: list of all applicants per posting
- Candidate cards: profile summary, qualification match score, location distance
- Stage management: move candidates through pipeline stages
- Interview scheduler: propose slots, send automated reminders
- Offer letter generation: pull from NAPS-compliant template, fill details, e-sign, send

Active Apprentice Management

- Table of all active apprentices: name, trade, location, start date, end date
- Attendance marking interface: supervisor marks daily attendance (bulk or individual)
- Training milestone logging: mark modules/skills completed
- Mid-term and end-term performance review forms
- Alert when any apprentice has unexplained absence for 3+ consecutive days

NAPS Compliance Centre

- Registration status per apprentice: Registered / Pending / Error
- Document status tracker: what has been submitted, what is pending, what is expiring
- Stipend payment confirmation upload (or payment initiation via platform)
- Government subsidy claim status: Filed / Under Review / Approved / Disbursed
- Compliance score (0–100) with improvement recommendations

ESG & Impact Reporting

- Auto-generated ESG report: women hired, sectors, stipends paid, certificates issued, retention rate
- Download as PDF or Excel
- Time period selector (quarterly, annual, custom)
- Benchmark comparison: how employer compares to industry average on gender diversity

admin use case

ID	Use Case	Actor	Description
UC-A01	Master Dashboard	Admin	Single-screen overview of all platform KPIs: registrations, active placements, compliance status, pending actions, grievances, stipend processing.
UC-A02	Employer Verification & Onboarding	Admin	Reviews employer registration, validates credentials, approves or rejects. Sends onboarding kit.
UC-A03	Job Posting Review & Approval	Admin	Reviews employer job postings for NAPS compliance, safety standards, and wage compliance before publishing.
UC-A04	NAPS Filing Management	Admin	Files and tracks all NAPS portal registrations on behalf of employer partners. Manages bulk filing, status tracking, and error resolution.
UC-A05	Stipend Compliance Monitoring	Admin	Monitors whether employers are paying NAPS-mandated minimum stipends. Flags and escalates non-compliance.
UC-A06	Grievance Management	Admin	Receives, and resolves all candidate grievances. Escalates safety issues to emergency protocol.
UC-A07	Candidate Verification	Admin	Reviews uploaded documents (Aadhaar, education certificates) and approves candidate profiles for placement.
UC-A08	Placement Matching	Admin	Algorithmically matches verified candidates to appropriate job postings based on qualification, location, and trade.
UC-A09	Analytics & Impact Reporting	Admin	Generates detailed impact reports for government bodies, CSR donors, and DFIs — covering placements, sectors, stipends, retention, and completion.
UC-A10	Subsidy Claim Filing	Admin	Prepares and files NAPS government reimbursement claims for employer partners. Tracks approval and disbursement.
UC-A11	Content Management	Admin	Manages multilingual content: job listings, financial literacy modules, community stories, FAQs, and notifications.
UC-A12	User & Role Management	Admin	Creates and manages user accounts, roles, and permissions for all stakeholder types including sub-admin roles for Even Cargo field staff.
UC-A13	Audit Trail & Compliance Log	Admin	Full audit trail of all platform actions — document uploads, status changes, payments — for regulatory compliance and dispute resolution.

Admin Portal Pages

Master Operations Dashboard

- Real-time KPIs: total registrations, active placements, NAPS filings, grievances open, stipends processed
- Geographic heatmap: concentration of candidates and placements by state and district
- Funnel chart: registration → profile complete → applied → shortlisted → placed → active → certified

- Priority inbox: pending approvals, open grievances, overdue SLAs

Candidate Management

- Full candidate database with advanced search and filter
- Document verification queue: review and approve uploaded documents
- Candidate profile edit and override capability
- Placement matching: suggest or manually assign candidate to employer listing
- Bulk status update and communication tools

Employer Management

- Employer onboarding queue: review registrations, verify credentials, approve/reject
- Employer directory with compliance score, active apprentice count, open grievances
- Employer profile edit and capability to issue compliance warnings
- Employer suspension / deactivation flow with audit log

NAPS Operations Centre

- NAPS filing queue: all pending filings by employer and apprentice
- Bulk NAPS registration tool: submit multiple apprentice registrations in one session
- Error tracker: NAPS portal rejection reasons and resolution workflow
- Government subsidy claim filing and tracking dashboard
- Deadline calendar: all NAPS filing deadlines across all employer accounts

Grievance Command Centre

- All grievances across all candidates: open, in-progress, resolved, escalated
- SLA tracker: time since submission, time to resolution, SLA breach alerts
- Grievance assignment to case handlers
- Safety escalation protocol: automatic trigger for harassment/safety grievances
- Resolution workflow with structured notes and evidence upload

Analytics & Impact Reporting

- Custom report builder: select dimensions (geography, sector, employer, trade, time period) and metrics
- Pre-built reports: NAPS Summary, Gender Inclusion Impact, Employer Performance, Facilitator Performance
- Export to PDF, Excel, PowerPoint
- Scheduled report delivery via email to government partners and CSR donors

Content Management System

- Manage all multilingual content: job listing review, financial literacy modules, community stories
- Push notifications and SMS/WhatsApp message composer
- Chatbot response management and FAQ updates
- Announcement publisher (platform-wide or targeted by user type/geography)

integrations

Non-Functional Requirements

Safety & Privacy

- Candidate location data: never shared with employers; used only for matching and SOS
- Aadhaar data: masked in all displays (last 4 digits only); full data encrypted at rest
- Grievance data: accessible only to Even Cargo safety team; never visible to employer
- HTTPS everywhere; end-to-end encryption for sensitive document uploads

Language & Localisation

- UI available in: English, Hindi, Tamil, Telugu, Kannada, Bengali, Marathi
- All notifications (SMS, WhatsApp, push) delivered in user's preferred language
- All PDFs (offer letters, certificates) generated in English + user's preferred language

Third-Party Integrations

Integration	Purpose	Priority
NAPS Portal (Apprenticeship India)	All NAPS registrations, filings, and status sync	P0 — Critical
Aadhaar / DigiLocker	Identity verification and document pull	P0 — Critical
ESIC & EPFO APIs	Social security enrolment for apprentices	P0 — Critical
Payment Gateway (Razorpay / PayU)	Stipend disbursement processing	P0 — Critical
MSG91	SMS notifications in multiple languages	P0 — Critical
WhatsApp Business API	WhatsApp-based registration, updates, and support	P1 — High
Google Maps	Location-based job discovery and workplace mapping	P1 — High
DocuSign / Leegality	Digital e-signing of offer letters and contracts	P1 — High
AWS Rekognition / OCR	Automated document reading and form auto-fill	P1 — High
Google Analytics / Mixpanel	User behaviour analytics and funnel tracking	P2 — Medium

Core Data Entities

The following are the primary data objects that drive the portal. Each entity maps to a database table and a set of API endpoints.

Entity	Key Fields	Relationships
Candidate	ID, name, DOB, mobile, aadhaar (encrypted), education, bank	Has many: Applications, Stipend Records, Grievances, Training Records

Entity	Key Fields	Relationships
	account, location, language, NAPS ID, status	
Employer	ID, company name, CIN, GST, sector, HQ address, authorized signatory, safety score, NAPS establishment ID, status	Has many: Job Postings, Active Apprentices, NAPS Filings
Job Posting	ID, employer ID, trade, qualification required, stipend amount, location, seats, duration, NAPS trade code, status	Belongs to: Employer; Has many: Applications
Application	ID, candidate ID, job posting ID, status, applied date, interview date, offer date, rejection reason	Belongs to: Candidate, Job Posting
Apprenticeship Contract	ID, candidate ID, employer ID, job posting ID, start date, end date, trade, stipend, NAPS registration status	Has many: Attendance Records, Stipend Records, Assessments
Attendance Record	ID, contract ID, date, status (present/absent/half-day), marked by, remarks	Belongs to: Apprenticeship Contract
Stipend Record	ID, contract ID, month, amount, due date, payment date, payment status, transaction reference	Belongs to: Apprenticeship Contract
NAPS Filing	ID, employer ID, contract ID, filing type, submission date, NAPS reference number, status, error notes	Belongs to: Employer, Apprenticeship Contract
Grievance	ID, candidate ID, employer ID, category, description, severity, status, handler ID, created date, resolved date	Belongs to: Candidate; optionally Employer
Subsidy Claim	ID, employer ID, filing period, number of apprentices, total claim amount, submission date, status, disbursement date	Belongs to: Employer

workflow+ recommended stack

Recommended Stack

Frontend

- React / Next.js
- PWA-enabled
- Tailwind CSS
- Mobile-first responsive UI

Backend

- Node.js + Express / NestJS
- REST APIs
- JWT Authentication
- Role-based access control

Database

- PostgreSQL

Storage

- AWS S3 / equivalent for document uploads

Infrastructure

- AWS / Azure
- CDN enabled
- HTTPS enforced

Notifications

- SMS
- WhatsApp
- Push notifications
- Email

Key Workflows

Candidate Journey

Registration → Profile Completion → Document Upload → Job Discovery → Apply → Interview → Offer
→ NAPS Registration → Active Apprenticeship → Training → Stipend Tracking → Completion

Employer Journey

Register → Verification → Post Jobs → Review Candidates → Interviews → Offer → Manage
Apprentices → Compliance → Subsidy Claims

Admin Journey

Verify → Approve → Match → File NAPS → Monitor Compliance → Resolve Grievances → Generate
Reports